

Crash Severity Prediction for Emergency Dispatch

Prediction Road Collision Severity from Scene Conditions Using Machine learning

Course Data Stewardship 2026 TU Wien

This experiment aims to predict how severe a road traffic accident is. It will be classified in Slight, Serious or Fatal. The Classification will be based on the conditions that are typically known at the moment of an emergency call. The goal is to help dispatch centres to decide what kind of emergency it is and what they should send: a police car, an ambulance, or a full trauma team.

The Data comes from the UK Road Safety Open Data (STATS19) for 2023, published by the UK Department for Transport on data.gov.uk under the Open Government Licence. The dataset is available at: “<https://www.gov.uk/government/statistical-data-sets/road-safety-open-data>”

It consists of three CSV files covering 104.258 collisions in the file `dft-road-casualty-statistics-collision-2023`, 189.815 vehicles in the file `dft-road-casualty-statistics-vehicle-2023` and 132.977 casualties in the file `dft-road-casualty-statistics-casualty-2023`. These three CSV files are linked together with a shared collision ID.

The features used as input for the model will be road type, speed limit, weather, lighting, road surface, time of day, day of the week, number of vehicles involved and vehicle type.

A Random Forest Classifier will be trained on 70% of the data, with 15% used for validation and 15% for testing.

The experiment produces the following output files: histogram plots for each feature during data exploration, a feature importance chart showing which conditions most influence severity, a confusion matrix visualising model performance across the three classes, a classification report (precision, recall, F1-score), a CSV file with predicted vs. actual labels on the test set, and a saved Random Forest model file (.pkl).

Requirements

Requirement	Where it's addressed
Existing Reusable data	STATS19 2023 — published by UK Dept. for Transport.
Permitted data source	data.gov.uk — UK national open data portal (permitted as 'other national portals').

Data Source URL	https://www.gov.uk/government/statistical-data-sets/road-safety-open-data
Licence	Open Government Licence v3.0 — free reuse with attribution.
Licence URL	https://www.nationalarchives.gov.uk/doc/open-government-licence/version/3/
Train / Validation / Test split	70% training / 15% validation / 15% test
ML model as output	Random Forest Classifier serialised as a .pkl file via scikit-learn.
Output files	Histogram plots for each feature (PNG), a feature importance chart (PNG), a confusion matrix (PNG), a classification report (precision, recall, F1-score) (CSV), a CSV file with predicted vs. actual labels on the test set, and a saved Random Forest model file (.pkl).
Implementation	Python: pandas (data prep), scikit-learn (ML), matplotlib/seaborn (visualisation).

